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Connected Cardiovascular Intelligence

A Framework for Longitudinal Multimodal Cardiovascular Data Integration

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Publication Type

Scientific and Technical White Paper

Keywords

Connected Cardiovascular Intelligence; cardiovascular medicine; longitudinal cardiovascular data; multimodal data; cardiovascular artificial intelligence; machine learning; clinical decision support; cardiovascular imaging; electrocardiography; digital health; interoperability; FHIR; DICOM; wearable technology; remote patient monitoring; patient-state modeling; clinical AI governance; AI validation; human-AI interaction; cardiovascular informatics. [1,23]

Conflict of Interest Disclosure

The author is associated with xBxBio, the organization developing and proposing the Connected Cardiovascular Intelligence framework described in this white paper. This relationship should be considered when interpreting the concepts and propositions presented herein.

No claim of independent external validation is made unless explicitly identified and supported by cited evidence.

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AI-Assisted Development Disclosure

Artificial intelligence tools were used to assist with the development, organization, drafting, and refinement of portions of this white paper.

The author retains responsibility for reviewing the manuscript, verifying references and factual claims, evaluating scientific interpretations, resolving errors, and approving the final published content.

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This white paper presents a proposed scientific, clinical, and technical framework for Connected Cardiovascular Intelligence.

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Version History

Version	Status	Year	Description
0.1	Scientific Review Draft	2026	Initial consolidated framework manuscript prepared for scientific, clinical, technical, quality, regulatory, and reference review.
0.2	Evidence-Mapped Scientific Review Draft	2026	Verified-source mapping and citation insertion; regulatory/standards references updated; CCI propositions retained as proposed/testable framework statements.

Version	Status	Year	Description
0.3	Publication- Readiness Review Draft	2026	Scientific consistency, citation coverage, current FDA/QMSR/CSA/cybersecurity updates, proposition register, heading renumbering, and publication-readiness controls.
0.4	Bibliographical y Verified External Review Draft	2026	Full bibliographic verification completed; duplicate reference removed; citation numbering reconciled; dynamic 2026 regulatory sources re-verified before external review.
0.5-P	Public Scientific Release	2026	Controlled public release for scientific discussion; Zenodo DOI assigned; external scientific review remains pending; not journal peer reviewed; public IP boundary applied.

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Table of Contents

1. Abstract
2. Introduction
3. Fragmentation of Cardiovascular Information
4. Limitations of Snapshot-Based Interpretation
5. Definition of Connected Cardiovascular Intelligence
6. Longitudinal Cardiovascular Patient State
7. Multimodal Cardiovascular Data Domains
8. Interoperability and Semantic Architecture
9. Temporal Data Architecture
10. Artificial Intelligence, Machine Learning, and Computational Intelligence
11. Clinical Decision-Support Layer
12. Human Clinical Oversight
13. Validation and Evidence
14. Quality and Regulatory Governance

15. Privacy, Security, and Data Governance

16. Proposed xBxBio Connected Cardiovascular Intelligence Framework

17. Research Agenda

18. Limitations

19. Conclusions

20. References

Proposition	Name	Publication Status
1	Longitudinal Context	Proposed/testable xBxBio framework proposition; not demonstrated clinical superiority.
2	Connected-State	Proposed/testable xBxBio framework proposition; not demonstrated clinical superiority.
3	Dynamic Patient-State	Proposed/testable xBxBio framework proposition; not demonstrated clinical superiority.
4	Multimodal Complementarity	Proposed/testable xBxBio framework proposition; not demonstrated clinical superiority.
5	Semantic Continuity	Proposed/testable xBxBio framework proposition; not demonstrated clinical superiority.
6	Temporal Coherence	Proposed/testable xBxBio framework proposition; not demonstrated clinical superiority.
7	Computational Augmentation	Proposed/testable xBxBio framework proposition; not demonstrated clinical

Proposition	Name	Publication Status
		superiority.
8	Clinician-Centered Decision Support	Proposed/testable xBxBio framework proposition; not demonstrated clinical superiority.
9	Human Clinical Authority	Proposed/testable xBxBio framework proposition; not demonstrated clinical superiority.
10	Evidence Proportionality	Proposed/testable xBxBio framework proposition; not demonstrated clinical superiority.
11	Governed Lifecycle	Proposed/testable xBxBio framework proposition; not demonstrated clinical superiority.
12	Trustworthy Data Stewardship	Proposed/testable xBxBio framework proposition; not demonstrated clinical superiority.
13	Closed-Loop Intelligence	Proposed/testable xBxBio framework proposition; not demonstrated clinical superiority.
14	Progressive Evidence Generation	Proposed/testable xBxBio framework proposition; not demonstrated clinical superiority.
15	Bounded Intelligence	Proposed/testable xBxBio framework proposition; not demonstrated clinical superiority.

About xBxBio

xBxBio advances Connected Cardiovascular Intelligence through responsible integration of longitudinal cardiovascular information, multimodal clinical data, interoperability, clinical artificial intelligence, data integrity, provenance, governance, validation, and human clinical oversight.

xBxBio Official Website

<https://xbxbio.com>

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